

TOOLS AND TECHNOLOGIES

The success of **Shifty** relies on the effective use of modern tools, frameworks, and technologies. The choice of technologies is based on **performance, scalability, and developer productivity**.

6.1 Programming Languages

1. JavaScript / TypeScript

- Core language for **React Native** mobile app and Node.js backend.
- TypeScript ensures **type safety and fewer runtime errors**.

2. Python

- Used in **machine learning modules** (Random Forest, XGBoost).
- Integration with Flask/FastAPI for deploying ML models as APIs.

3. SQL (for Analytics)

- Although MongoDB is the primary database, **SQL** will be used for **data analysis and reporting**.

6.2 Frameworks and Libraries

1. Frontend Framework

- **React Native**: Enables cross-platform mobile development with a single codebase.
- **Redux Toolkit**: For state management.
- **Axios**: For API requests.

2. Backend Framework

- **Node.js with Express.js**: Lightweight, scalable backend service.
- **FastAPI/Flask**: Hosting ML models with REST APIs.